Summary Statistics by Metro x Trip Category:

Core City	TripCategory	AvgDuration	StdDevDuration	MinDuration	Q1_Approx	Median_Approx	Q3_Approx	MaxDuration
Austin	Commuter	7.14	3.43	1	5	7	10	17.5
Austin	Leisure	17.46	14.87	0	7	14	30	64.5
Los Angeles	Commuter	9.19	5.3	1	5	8	13	25
Los Angeles	Leisure	15.97	12.59	0	7	14	28	59.5
Pittsburgh	Commuter	11.73	6.32	1	7	11	17	32
Pittsburgh	Leisure	30.1	28.3	0	10	22	56	125
Portland	Commuter	10.09	6.12	1	6	9	15	28.5
Portland	Leisure	17.52	14.48	1	7	14	30	64.5

Share of Trip Types (Short, Typical, Long) Within Each Metro and Category:

MetroArea	TripCategory	DurationBand	TripCount	PctWithinCategory
Austin	Commuter	Shorter Trips (Bottom 25% of Metro)	81227	50.64
Austin	Commuter	Typical Trips (Middle 50% of Metro)	79170	49.36
Austin	Leisure	Shorter Trips (Bottom 25% of Metro)	265128	26.78
Austin	Leisure	Typical Trips (Middle 50% of Metro)	447015	45.14
Austin	Leisure	Longer Trips (Top 25% of Metro)	278041	28.08
Los Angeles	Commuter	Shorter Trips (Bottom 25% of Metro)	83710	38.61
Los Angeles	Commuter	Typical Trips (Middle 50% of Metro)	120070	55.38

Los Angeles	Commuter	Longer Trips (Top 25% of Metro)	13029	6.01
Los Angeles	Leisure	Shorter Trips (Bottom 25% of Metro)	182709	24.52
Los Angeles	Leisure	Typical Trips (Middle 50% of Metro)	338876	45.48
Los Angeles	Leisure	Longer Trips (Top 25% of Metro)	223569	30
Pittsburgh	Commuter	Shorter Trips (Bottom 25% of Metro)	20908	44.38
Pittsburgh	Commuter	Typical Trips (Middle 50% of Metro)	26206	55.62
Pittsburgh	Leisure	Shorter Trips (Bottom 25% of Metro)	104449	26.32
Pittsburgh	Leisure	Typical Trips (Middle 50% of Metro)	181551	45.76
Pittsburgh	Leisure	Longer Trips (Top 25% of Metro)	110776	27.92
Portland	Commuter	Shorter Trips (Bottom 25% of Metro)	49828	34.73
Portland	Commuter	Typical Trips (Middle 50% of Metro)	86405	60.22
Portland	Commuter	Longer Trips (Top 25% of Metro)	7246	5.05
Portland	Leisure	Shorter Trips (Bottom 25% of Metro)	249844	23.99
Portland	Leisure	Typical Trips (Middle 50% of Metro)	504254	48.43
Portland	Leisure	Longer Trips (Top 25% of Metro)	287177	27.58

Dockless vs. Trip Duration by Metro & Category:

MetroArea	TripCategory	TripMode	TripCount	AvgDuration_Min	StdDevDuration_Min
Austin	Commuter	Station-to-Station	160397	7.14	3.43
Austin	Leisure	Station-to-Station	990184	17.46	14.87

Los Angeles	Commuter	Station-to-Station	216809	9.19	5.3
Los Angeles	Leisure	Station-to-Station	745154	15.97	12.59
Pittsburgh	Commuter	Dockless Involved	4135	11.39	6.45
Pittsburgh	Commuter	Station-to-Station	42979	11.76	6.31
Pittsburgh	Leisure	Dockless Involved	30115	26.73	26.33
Pittsburgh	Leisure	Station-to-Station	366661	30.38	28.44
Portland	Commuter	Dockless Involved	53579	9.6	6.19
Portland	Commuter	Station-to-Station	89900	10.38	6.05
Portland	Leisure	Dockless Involved	480442	15.3	13.52
Portland	Leisure	Station-to-Station	560833	19.42	15

Average Trip Distance by Metro and Category:

MetroArea	TripCategory	Trips	AvgTripDistance_Miles	StdDevTripDistance_Miles
Austin	Commuter	160137	0.66	0.34
Austin	Leisure	985282	0.68	0.54
Los Angeles	Commuter	209156	0.78	0.48
Los Angeles	Leisure	696645	0.69	0.6
Pittsburgh	Commuter	42394	1.13	0.71
Pittsburgh	Leisure	361459	0.85	0.91
Portland	Commuter	89900	0.64	0.5
Portland	Leisure	560833	0.48	0.47

Distance + Duration Efficiency:

MetroArea	TripCategory	AvgTripDistance_Miles	AvgTripDuration_Min	AvgMilesPerMinute
Austin	Commuter	0.66	7.14	0.0966
Austin	Leisure	0.68	17.46	0.0583
Los Angeles	Commuter	0.78	9.07	0.0929
Los Angeles	Leisure	0.69	15.69	0.0624
Pittsburgh	Commuter	1.13	11.76	0.1018
Pittsburgh	Leisure	0.85	30.43	0.055
Portland	Commuter	0.64	10.38	0.0731
Portland	Leisure	0.48	19.42	0.0399

SQL Code:

```
DROP TEMPORARY TABLE IF EXISTS ClassifiedTrips;

CREATE TEMPORARY TABLE ClassifiedTrips AS
SELECT
   TripID,
   MetroID,
   StartDate,
   StartTime,
   StartDayOfWeek,
   FromStationID,
   ToStationID,
   UserType,
   TripDuration_Minutes,
   CASE
      WHEN StartDayOfWeek BETWEEN 1 AND 5
          AND UserType = 'Subscriber'
          AND (
              (StartTime BETWEEN '06:30:00' AND '09:30:00')
              OR (StartTime BETWEEN '16:00:00' AND '19:00:00')
             )
          AND FromStationID <> ToStationID
      THEN 'Commuter'
      ELSE 'Leisure'
   END AS TripCategory
FROM Rentals
WHERE NOT (
     TripDuration_Minutes = 0
     AND FromStationID <> ToStationID
  );

--Compute Quartiles into a Temporary Table
DROP TEMPORARY TABLE IF EXISTS TripQuartiles;

CREATE TEMPORARY TABLE TripQuartiles AS
WITH RankedDurations AS (
   SELECT
      MetroID,
      TripCategory,
      TripDuration_Minutes,
      PERCENT_RANK() OVER (
         PARTITION BY MetroID, TripCategory
         ORDER BY TripDuration_Minutes
```

```
        ) AS pr
    FROM ClassifiedTrips
)
SELECT
    MetroID,
    TripCategory,
    MAX(CASE WHEN pr <= 0.25 THEN TripDuration_Minutes END) AS Q1,
    MAX(CASE WHEN pr <= 0.75 THEN TripDuration_Minutes END) AS Q3
FROM RankedDurations
GROUP BY MetroID, TripCategory;

--Create Clean Trips (without outliers) Using That Table
DROP TEMPORARY TABLE IF EXISTS ClassifiedTrips_Clean;

CREATE TEMPORARY TABLE ClassifiedTrips_Clean AS
SELECT C.*
FROM ClassifiedTrips C
JOIN TripQuartiles Q
  ON C.MetroID = Q.MetroID
 AND C.TripCategory = Q.TripCategory
WHERE C.TripDuration_Minutes BETWEEN
      (Q.Q1 - 1.5 * (Q.Q3 - Q.Q1))
  AND (Q.Q3 + 1.5 * (Q.Q3 - Q.Q1));

--Diagnostic Check for How Many Outliers Removed
SELECT
    M.CoreCity,
    C.TripCategory,
    COUNT(*) AS OriginalTrips,
    SUM(CASE WHEN CL.TripID IS NULL THEN 1 ELSE 0 END) AS RemovedAsOutlier,
    ROUND(
        100.0 * SUM(CASE WHEN CL.TripID IS NULL THEN 1 ELSE 0 END) / COUNT(*),
        2
    ) AS PctRemoved
FROM ClassifiedTrips C
JOIN MetroArea M ON C.MetroID = M.MetroID
LEFT JOIN ClassifiedTrips_Clean CL
  ON C.TripID = CL.TripID
GROUP BY M.CoreCity, C.TripCategory;

--1) Retrive Table With All Trips to Create Box Plot
SELECT
    C.TripID,
    M.CoreCity,
```

```
    C.TripDuration_Minutes,
    C.TripCategory
FROM ClassifiedTrips C
JOIN MetroArea M ON C.MetroID = M.MetroID;

-- 2) Summary Statistics by Metro x Trip Category

DROP TEMPORARY TABLE IF EXISTS ClassifiedTrips_Ranked;

CREATE TEMPORARY TABLE ClassifiedTrips_Ranked AS
SELECT
    TripID,
    MetroID,
    TripCategory,
    TripDuration_Minutes,
    PERCENT_RANK() OVER (
        PARTITION BY MetroID, TripCategory
        ORDER BY TripDuration_Minutes
    ) AS pr
FROM ClassifiedTrips;

WITH Statistics_Clean AS (
    SELECT
        MetroID,
        TripCategory,
        MIN(TripDuration_Minutes) AS MinDuration,
        MAX(CASE WHEN pr <= 0.25 THEN TripDuration_Minutes END) AS Q1_Approx,
        MAX(CASE WHEN pr <= 0.50 THEN TripDuration_Minutes END) AS Median_Approx,
        MAX(CASE WHEN pr <= 0.75 THEN TripDuration_Minutes END) AS Q3_Approx,
        MAX(TripDuration_Minutes) AS MaxDuration
    FROM ClassifiedTrips_Ranked
    GROUP BY MetroID, TripCategory
)
SELECT
    M.CoreCity,
    C.TripCategory,
    ROUND(AVG(C.TripDuration_Minutes), 2) AS AvgDuration,
    ROUND(STDDEV(C.TripDuration_Minutes), 2) AS StdDevDuration,
    S.MinDuration,
    S.Q1_Approx,
    S.Median_Approx,
    S.Q3_Approx,
    S.MaxDuration
FROM ClassifiedTrips_Clean AS C
```

```
JOIN MetroArea M
   ON C.MetroID = M.MetroID
JOIN Statistics_Clean S
   ON C.MetroID = S.MetroID
  AND C.TripCategory = S.TripCategory
GROUP BY
   M.CoreCity,
   C.TripCategory
ORDER BY M.CoreCity, C.TripCategory;

--3) Share of Trip Types (Short, Typical, Long) Within Each Metro and Category
WITH MetroRankedDurations AS (
   SELECT
      MetroID,
      TripID,
      TripCategory,
      TripDuration_Minutes,
      PERCENT_RANK() OVER (
         PARTITION BY MetroID
         ORDER BY TripDuration_Minutes
      ) AS pr
   FROM ClassifiedTrips_Clean
)
SELECT
   M.CoreCity AS MetroArea,
   R.TripCategory,
   CASE
      WHEN R.pr <= 0.25 THEN 'Shorter Trips (Bottom 25% of Metro)'
      WHEN R.pr <= 0.75 THEN 'Typical Trips (Middle 50% of Metro)'
      ELSE 'Longer Trips (Top 25% of Metro)'
   END AS DurationBand,
   COUNT(*) AS TripCount,
   ROUND(
      100.0 * COUNT(*) /
      SUM(COUNT(*)) OVER (PARTITION BY M.CoreCity, R.TripCategory),
      2
   ) AS PctWithinCategory
FROM MetroRankedDurations R
JOIN MetroArea M ON R.MetroID = M.MetroID
GROUP BY M.CoreCity, R.TripCategory, DurationBand
ORDER BY M.CoreCity, R.TripCategory,
       FIELD(DurationBand,
            'Shorter Trips (Bottom 25% of Metro)',
            'Typical Trips (Middle 50% of Metro)',
```

```
            'Longer Trips (Top 25% of Metro)');

--4) Rentals beginning Dockless (not at a station) vs. Trip Duration
SELECT
   M.CoreCity AS MetroArea,
   C.TripCategory,
   CASE
      WHEN C.FromStationID = -5000 OR C.ToStationID = -5000 THEN 'Dockless Involved'
      ELSE 'Station-to-Station'
   END AS TripMode,
   COUNT(*) AS TripCount,
   ROUND(AVG(C.TripDuration_Minutes), 2) AS AvgDuration_Min,
   ROUND(STDDEV(C.TripDuration_Minutes), 2) AS StdDevDuration_Min
FROM ClassifiedTrips_Clean C
JOIN MetroArea M ON C.MetroID = M.MetroID
GROUP BY M.CoreCity, C.TripCategory, TripMode
ORDER BY M.CoreCity, C.TripCategory, TripMode;

--Base Table for Distance Analysis
DROP TEMPORARY TABLE IF EXISTS SpatialTrips;

CREATE TEMPORARY TABLE SpatialTrips AS
SELECT
   C.TripID,
   C.MetroID,
   C.TripCategory,
   C.TripDuration_Minutes,

   FS.StationName AS FromStationName,
   TS.StationName AS ToStationName,

   FS.Latitude AS FromLat,
   FS.Longitude AS FromLon,
   TS.Latitude AS ToLat,
   TS.Longitude AS ToLon,

   -- Haversine formula (miles)
   3959 * 2 * ASIN(
      SQRT(
         POWER(SIN(RADIANS(TS.Latitude - FS.Latitude) / 2), 2) +
         COS(RADIANS(FS.Latitude)) * COS(RADIANS(TS.Latitude)) *
         POWER(SIN(RADIANS(TS.Longitude - FS.Longitude) / 2), 2)
      )
   ) AS TripDistance_Miles
```

```
FROM ClassifiedTrips_Clean C
JOIN Stations FS
  ON C.FromStationID = FS.StationID
 AND C.MetroID = FS.MetroID
JOIN Stations TS
  ON C.ToStationID = TS.StationID
 AND C.MetroID = TS.MetroID

-- Exclude dockless trips (no real station location)
WHERE
   C.FromStationID <> -5000
   AND C.ToStationID <> -5000
   AND FS.Latitude > 0
   AND FS.Longitude < 0
   AND TS.Latitude > 0
   AND TS.Longitude < 0;

-- 5) Avg Trip Distance by Category × Metro
SELECT
   M.CoreCity AS MetroArea,
   S.TripCategory,
   COUNT(*) AS Trips,
   ROUND(AVG(S.TripDistance_Miles), 2) AS AvgTripDistance_Miles,
   ROUND(STDDEV(S.TripDistance_Miles), 2) AS StdDevTripDistance_Miles
FROM SpatialTrips S
JOIN MetroArea M ON S.MetroID = M.MetroID
GROUP BY M.CoreCity, S.TripCategory
ORDER BY M.CoreCity, S.TripCategory;

-- 6) Distance + Duration Efficiency (Miles per Minute)
SELECT
   M.CoreCity AS MetroArea,
   S.TripCategory,
   ROUND(AVG(S.TripDistance_Miles), 2) AS AvgTripDistance_Miles,
   ROUND(AVG(S.TripDuration_Minutes), 2) AS AvgTripDuration_Min,
   ROUND(
      AVG(S.TripDistance_Miles / NULLIF(S.TripDuration_Minutes, 0)),
      4
   ) AS AvgMilesPerMinute
FROM SpatialTrips S
JOIN MetroArea M ON S.MetroID = M.MetroID
GROUP BY M.CoreCity, S.TripCategory
ORDER BY M.CoreCity, S.TripCategory;
```